

House Price Prediction Using Machine Learning:

Model Validation, Overfitting Control and Hyperparameter Tuning

1. Objective

The objective of this project was to improve the reliability and generalization capability of machine learning models by applying model validation techniques on the California Housing Dataset. The project focused on detecting overfitting, performing cross-validation, tuning model hyperparameters using GridSearchCV, and selecting the best-performing regression model based on quantitative evaluation metrics.

2. Dataset Description

The California Housing Dataset was obtained from the Scikit-Learn library. The target variable was **Median House Value (HousePrice)**, while the input features included:

- Median Income
- House Age
- Average Rooms
- Average Bedrooms
- Population
- Average Occupancy
- Latitude
- Longitude

The dataset contained **20,640 records** and **8 input features**.

3. Methodology

Data Preparation

The California Housing Dataset was loaded using Scikit-Learn and converted into a pandas DataFrame. The input features (X) and target variable (y) were separated before model training.

Feature Scaling

Since the numerical features were measured on different scales, StandardScaler was applied to standardize the feature values. Feature scaling improved model stability and ensured fair learning across all input variables.

Train-Test Split

The dataset was divided into training and testing sets using an **80:20 ratio** with a **random state of 42**. The training set was used to train the models, while the testing set was reserved for unbiased model evaluation.

Model Validation

An initial Decision Tree Regressor was trained to compare training and testing performance. A noticeable difference between training and testing errors indicated overfitting, demonstrating the need for model validation and complexity control.

Cross-Validation

Five-fold Cross Validation (5-Fold CV) was performed using **cross_val_score()** to obtain a more reliable estimate of model performance across multiple data splits instead of relying on a single train-test split.

Hyperparameter Tuning

GridSearchCV was used to optimize the Decision Tree Regressor by searching different combinations of **max_depth** and **min_samples_split**. The best-performing estimator was selected as the final model.

Machine Learning Models

Three regression algorithms were evaluated:

1. Linear Regression
2. Ridge Regression
3. Tuned Decision Tree Regressor (GridSearchCV)

4. Evaluation Metrics

Model performance was assessed using **Root Mean Squared Error (RMSE)** and **R² Score**. Lower RMSE values indicate better prediction accuracy, while higher R² values indicate stronger explanatory power. Cross-validation RMSE was also used to estimate model generalization performance.

5. Results and Performance Comparison

Model	RMSE	R ² Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Tuned Decision Tree Regressor	0.645430	0.682099

6. Best Model Selection

Among the evaluated models, the **Tuned Decision Tree Regressor** achieved the **lowest RMSE (0.645430)** and the **highest R² Score (0.682099)**. Hyperparameter tuning successfully reduced overfitting and improved the model’s ability to generalize to unseen data. Therefore, it was selected as the best-performing model for this dataset.

7. Model Validation

Cross-validation confirmed that the tuned Decision Tree produced stable performance across multiple folds, providing a more reliable estimate of model accuracy than a single train-test split. Comparing training and testing errors before and after hyperparameter tuning demonstrated that overfitting had been effectively controlled.

8. Conclusion

This project demonstrated advanced machine learning techniques for improving model reliability through model validation, cross-validation, and hyperparameter tuning. Three regression models were evaluated using the California Housing Dataset. Based on RMSE and R² Score, the Tuned Decision Tree Regressor achieved the best predictive performance.

The project highlights the importance of model validation in building robust machine learning solutions. Future improvements could include Random Forest, Gradient Boosting, XGBoost, and Bayesian optimization for hyperparameter tuning.

This matches the structure and writing style of your Task 2 report while updating the content for Task 3. If you want the final PDF to look almost identical—including the same font, margins, headings, and table layout—I can generate it in that exact style.